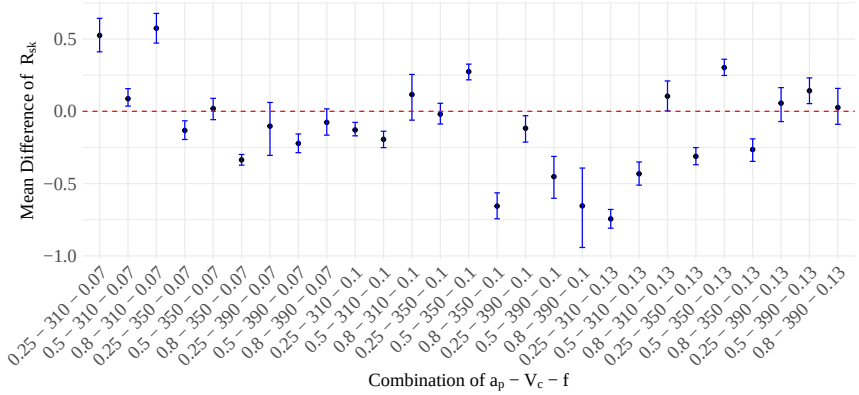
Probabilistic Spatial Roughness Metamodeling for Digital Roughness Mapping in the Non-Gaussian Descriptor Space

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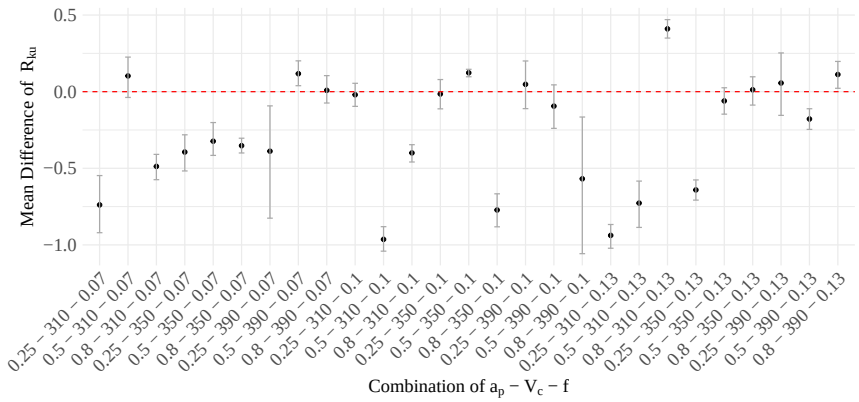
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Supplementary Materials

In the supplementary material, Figure S1 presents the bootstrap-based inter-trial comparison for additional roughness descriptors, including (a) R_{sk} and (b) R_{ku}. For each machining combination, 95% confidence intervals of the mean differences between Trial 1 and Trial 2 are estimated using 1000 bootstrap samples. The results are visualized using error bars, where intervals intersecting the zero reference line indicate no statistically significant difference, while non-intersecting intervals indicate statistically significant differences. These supplementary analyses complement the R_a results presented in the main text and further support the presence of inter-trial variability in the measured roughness parameters.



(a) R_{sk}



(b) R_{ku}

Figure S1: Bootstrap confidence intervals of Trial 1–Trial 2 mean differences for R_{sk} and R_{ku}. Error bars denote bootstrap confidence intervals; dashed line indicates zero difference.

Supplementary Table S1 provides the interval data derived from the ranges of Trial 1 and Trial 2 for each machining combination, which serve as inputs for estimating the first and second moment bounds used in the moment-bound-based MCS

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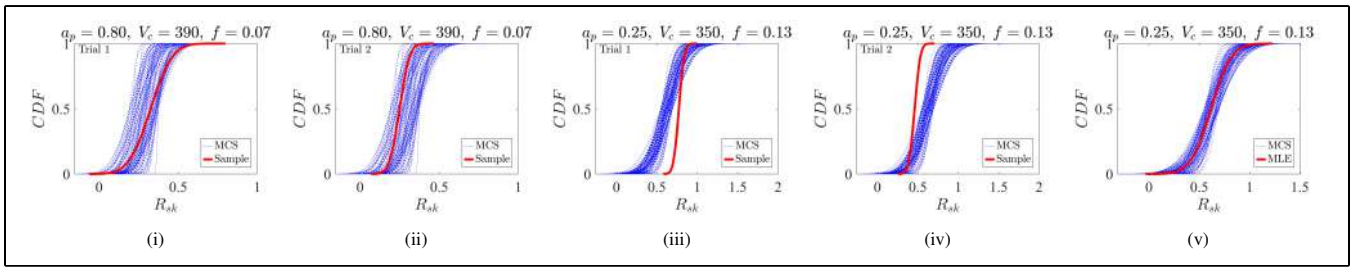
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framework. These interval representations capture the variability between repeated trials and form the basis for uncertainty propagation through simulation.

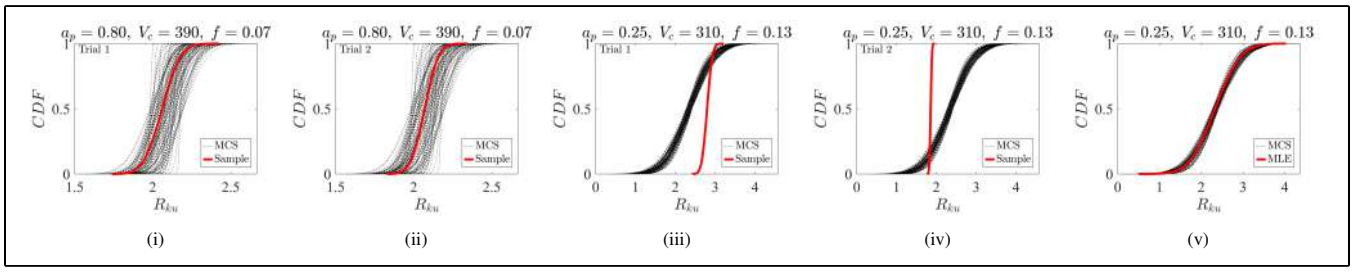
Table S1: Empirical multiple interval data of R_a, R_{sk}, and R_{ku} from the experimental runs of trial 1 and trial 2

Sl.	Combination			$R_a[lb, ub]$		$R_{sk}[lb, ub]$		$R_{ku}[lb, ub]$	
	a_p	V_c	f	Trial 1	Trial 2	Trial 1	Trial 2	Trial 1	Trial 2
1	0.25	310	0.07	[0.211, 0.264]	[0.378, 0.402]	$[-0.457, -0.201]$	[0.066, 0.444]	[2.638, 3.132]	[2.002, 2.374]
2	0.25	310	0.10	[0.667, 0.684]	[0.858, 0.882]	[0.084, 0.253]	[0.049, 0.094]	[1.494, 1.747]	[1.516, 1.641]
3	0.25	310	0.13	[0.798, 0.856]	[1.126, 1.18]	[0.975, 1.095]	[0.216, 0.373]	[2.691, 2.957]	[1.833, 1.892]
4	0.25	350	0.07	[0.359, 0.45]	[0.586, 0.677]	[0.530, 0.714]	[0.416, 0.540]	[2.229, 2.636]	[1.959, 2.097]
5	0.25	350	0.10	[0.612, 0.724]	[1.032, 1.062]	[0.571, 0.865]	[0.687, 0.773]	[2.148, 2.494]	[2.268, 2.373]
6	0.25	350	0.13	[1.125, 1.191]	[1.634, 1.812]	[0.701, 0.847]	[0.391, 0.549]	[2.384, 2.561]	[1.784, 1.927]
7	0.25	390	0.07	[0.293, 0.362]	[0.453, 0.49]	[0.289, 1.018]	[0.333, 0.532]	[1.824, 3.222]	[1.737, 1.867]
8	0.25	390	0.10	[0.692, 0.735]	[0.794, 1.019]	[0.512, 0.612]	[0.291, 0.583]	[1.872, 2.007]	[1.752, 2.228]
9	0.25	390	0.13	[1.07, 1.149]	[0.988, 1.551]	[0.422, 0.576]	[0.390, 0.724]	[2.028, 2.151]	[1.873, 2.407]
10	0.50	310	0.07	[0.243, 0.303]	[0.442, 0.456]	$[-0.048, 0.160]$	[0.144, 0.236]	[1.727, 2.217]	[1.949, 2.115]
11	0.50	310	0.10	[0.501, 0.538]	[0.888, 0.944]	[0.794, 0.947]	[0.578, 0.743]	[2.954, 3.197]	[2.032, 2.171]
12	0.50	310	0.13	[1.046, 1.133]	[0.952, 1.194]	[0.807, 1.052]	[0.428, 0.582]	[2.495, 3.055]	[1.945, 2.047]
13	0.50	350	0.07	[0.254, 0.356]	[0.362, 0.407]	$[-0.137, 0.076]$	$[-0.112, 0.058]$	[1.802, 2.165]	[1.615, 1.790]
14	0.50	350	0.10	[0.595, 0.623]	[0.605, 0.653]	[0.084, 0.181]	[0.310, 0.470]	[1.949, 1.997]	[2.050, 2.119]
15	0.50	350	0.13	[0.505, 0.573]	[0.96, 0.978]	[0.232, 0.431]	[0.625, 0.684]	[2.207, 2.488]	[2.244, 2.300]
16	0.50	390	0.07	[0.342, 0.371]	[0.255, 0.295]	[0.001, 0.111]	$[-0.256, -0.074]$	[1.831, 1.944]	[1.898, 2.183]
17	0.50	390	0.10	[0.532, 0.596]	[0.367, 0.418]	[0.381, 0.713]	$[-0.044, 0.245]$	[2.221, 2.628]	[2.184, 2.399]
18	0.50	390	0.13	[0.495, 0.541]	[0.658, 0.691]	[0.630, 0.808]	[0.716, 0.997]	[2.908, 3.136]	[2.768, 2.910]
19	0.80	310	0.07	[0.36, 0.404]	[0.603, 0.753]	$[-0.483, -0.269]$	[0.106, 0.373]	[2.075, 2.350]	[1.640, 1.828]
20	0.80	310	0.10	[0.586, 0.654]	[0.918, 1.014]	$[-0.039, 0.548]$	[0.127, 0.306]	[2.197, 2.323]	[1.753, 1.915]
21	0.80	310	0.13	[0.855, 1.213]	[0.924, 0.936]	[0.298, 0.585]	[0.477, 0.689]	[2.145, 2.297]	[2.579, 2.720]
22	0.80	350	0.07	[0.366, 0.383]	[0.539, 0.584]	[0.376, 0.463]	[0.051, 0.148]	[1.962, 2.107]	[1.637, 1.750]
23	0.80	350	0.10	[0.729, 0.929]	[0.644, 0.698]	[0.412, 0.708]	$[-0.122, -0.012]$	[2.312, 2.694]	[1.697, 1.738]
24	0.80	350	0.13	[1.382, 1.46]	[0.829, 0.894]	[0.809, 1.012]	[0.522, 0.694]	[2.528, 2.820]	[2.562, 2.753]
25	0.80	390	0.07	[0.344, 0.535]	[0.439, 0.489]	[0.171, 0.419]	[0.210, 0.343]	[1.957, 2.202]	[2.001, 2.202]
26	0.80	390	0.10	[0.449, 0.882]	[0.538, 0.61]	[0.081, 1.066]	$[-0.275, -0.013]$	[1.855, 3.403]	[1.711, 1.920]
27	0.80	390	0.13	[0.827, 1.198]	[0.821, 0.848]	[0.346, 0.703]	[0.453, 0.649]	[2.182, 2.446]	[2.328, 2.487]

Supplementary Figure S2 illustrates the comparison between sample-based empirical CDFs and MCS-generated simulation envelopes for additional roughness descriptors: (a) R_{sk} and (b) R_{ku}. For each selected machining condition, the empirical CDFs constructed from trial-wise sample statistics are evaluated against the corresponding simulation envelopes. The extent to which the sample-based empirical CDFs remain within or extend beyond these envelopes provides a visual and statistical indication of the presence or absence of significant inter-trial variability, complementing the results presented for R_a in the main text.



(a) R_{sk}



(b) R_{ku}

Figure S2: Empirical CDFs of R_{sk} and R_{ku} illustrating combination-wise trial-to-trial variability and MLE-based uncertainty quantification. Within each row, panels (i)–(ii) correspond to Trial 1 and Trial 2 for a machining condition exhibiting no statistically significant trial differences identified by bootstrap resampling, panels (iii)–(iv) show a condition with significant trial differences, and panel (v) presents the CDF constructed from MLE-estimated parameters.

Supplementary Figures S3–S5 present the comparison between CDFs constructed using MLE-based parameters and the corresponding moment-bound-based MCS simulation envelopes for R_a, R_{sk}, and R_{ku}, respectively, across representative machining combinations. The MLE-based CDFs are observed to lie consistently near the central region of the simulation envelopes, indicating strong agreement with the probabilistic bounds derived from the moment-bound framework. This consistency demonstrates that the MLE-based parameter estimates effectively capture the underlying distribution of the roughness parameters while remaining compatible with the uncertainty characterized through the MCS approach.

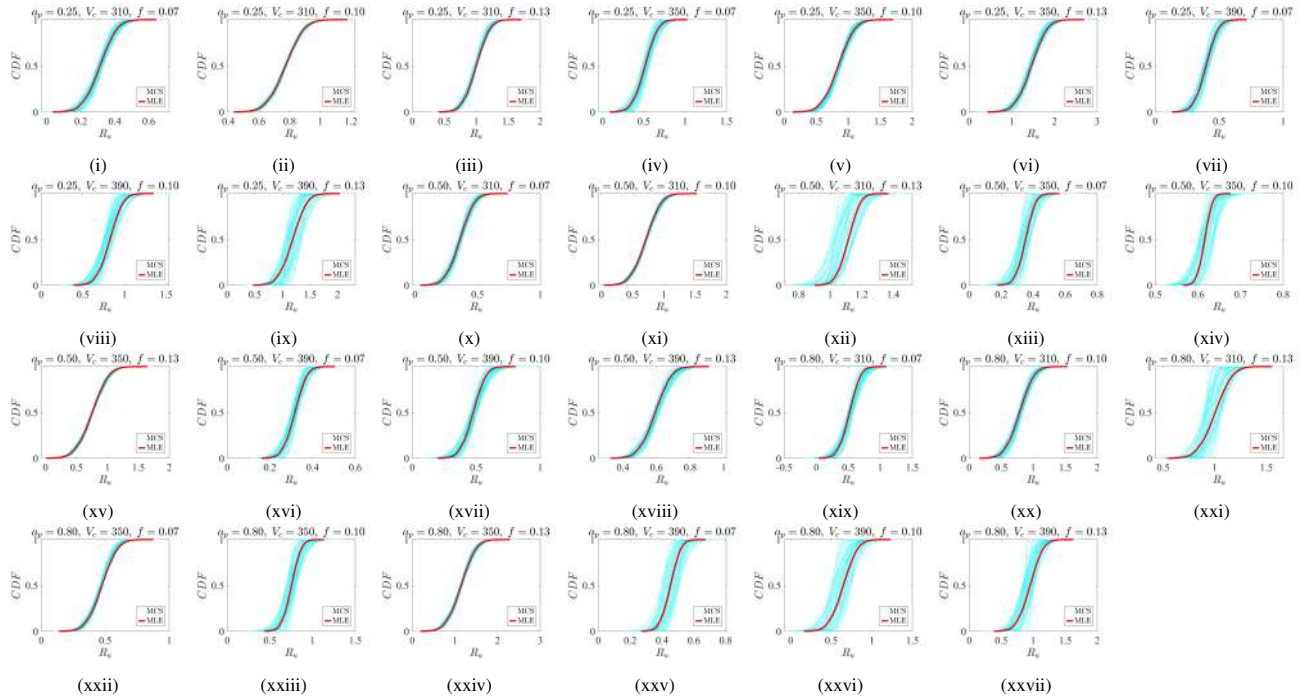


Figure S3: MLE-based and simulated CDFs of the kurtosis parameter R_a for all 27 machining combinations. Each panel compares the moment-bounded MCS envelope with the MLE-based CDF. The MCS envelope is generated by uniformly sampling admissible first- and second-moment bounds inferred from trial-wise interval uncertainty. For all machining combinations, the MLE-based CDF lies entirely within the MCS envelope, confirming consistency with the admissible moment space.

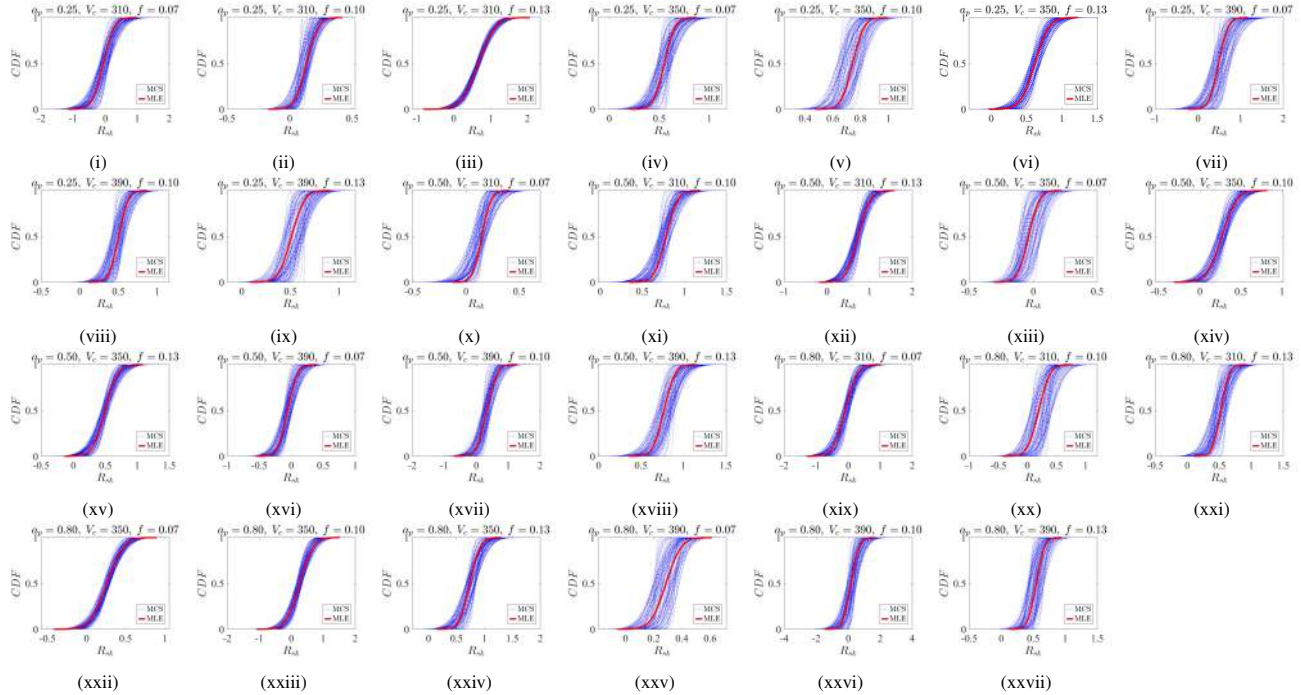


Figure S4: MLE-based and simulated CDFs of the kurtosis parameter R_{sk} for all 27 machining combinations. Each panel compares the moment-bounded MCS envelope with the MLE-based CDF. For all combinations, the MLE-derived CDF remains fully enclosed within the admissible MCS envelope, indicating consistency of the estimated skewness distributions with the epistemic uncertainty implied by trial-wise interval measurements.

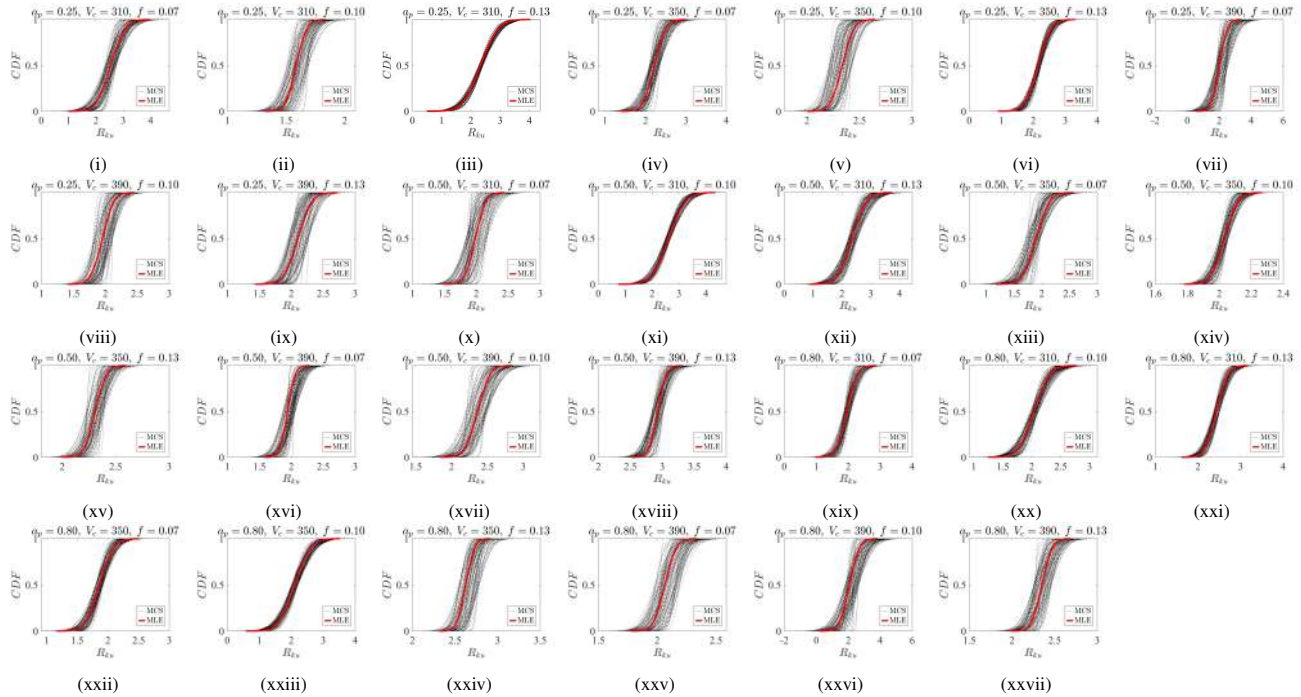
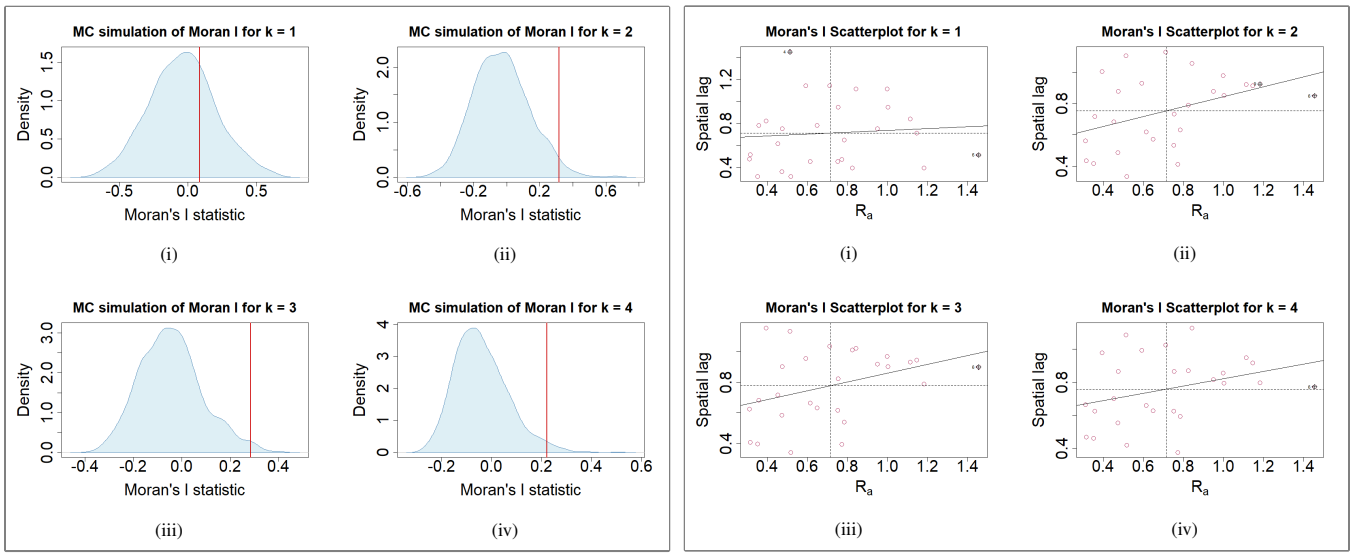


Figure S5: MLE-based and simulated CDFs of the kurtosis parameter R_{ku} for all 27 machining combinations. Each panel compares the moment-bounded MCS envelope with the MLE-based CDF. The complete enclosure of the MLE-based ECDF within the MCS envelope across all combinations demonstrates that the estimated tail behavior of surface roughness distributions is consistent with admissible second-moment uncertainty derived from interval measurements.

Supplementary Figure S6 provides visual diagnostics of the Global Moran's I analysis. Figure S6(a) presents the Monte Carlo simulation (MCS) distributions of Moran's I under the null hypothesis of spatial randomness for $k = 1$ to $k = 4$, with the observed statistics indicated by vertical lines. For $k = 2, 3$, and 4, the observed values lie in the upper tail of the distributions, supporting the presence of significant positive spatial autocorrelation. Figure S6(b) shows the corresponding Moran's I scatter plots, where the horizontal axis represents the observed R_a values and the vertical axis denotes the spatial lag. The positive slopes observed for $k = 2, 3$, and 4 further confirm the presence of spatial dependence, while also highlighting the influence of individual observations on the overall spatial structure.



(a) MCS-based Moran's I distributions **(b) Moran's I scatter plots**

Figure S6: Global Moran's I analysis for spatial autocorrelation in R_a: (a) Monte Carlo simulation (MCS)-based distributions of Moran's I, and (b) corresponding Moran's I scatter plots for $k = 1, 2, 3$, and 4.

Supplementary Figure S7 presents the universal kriging-based interpolation of average surface roughness R_a for all 27 machining combinations using MLE-based parameters. Figure S8 shows the corresponding kriging variance maps (σ_k^2), representing the spatial distribution of prediction uncertainty for each machining condition.

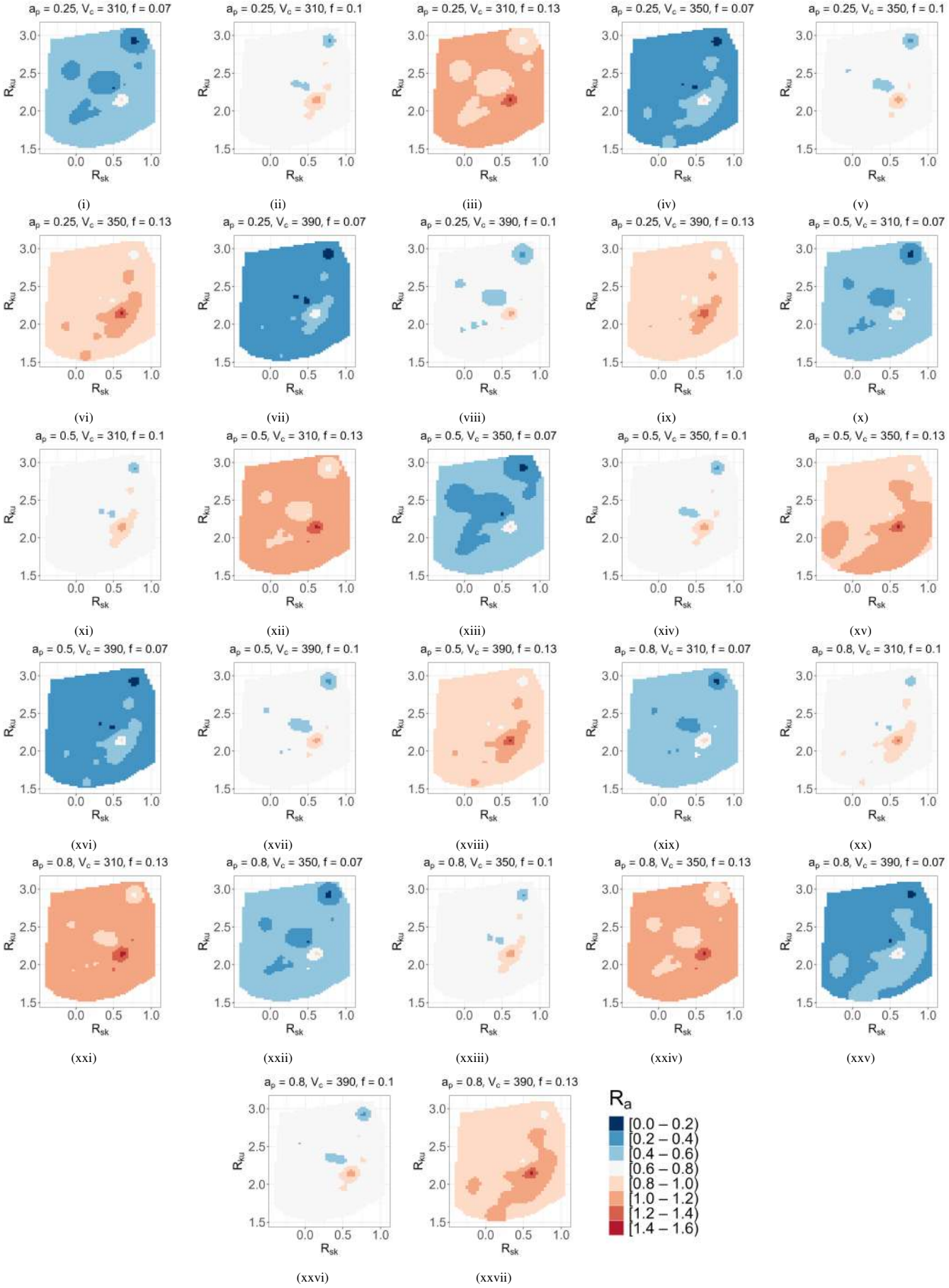


Figure S7: Spatial prediction of average surface roughness R_a across all 27 machining parameter combinations (a_p–V_c–f) using *MLE-based* universal kriging for digital roughness mapping.

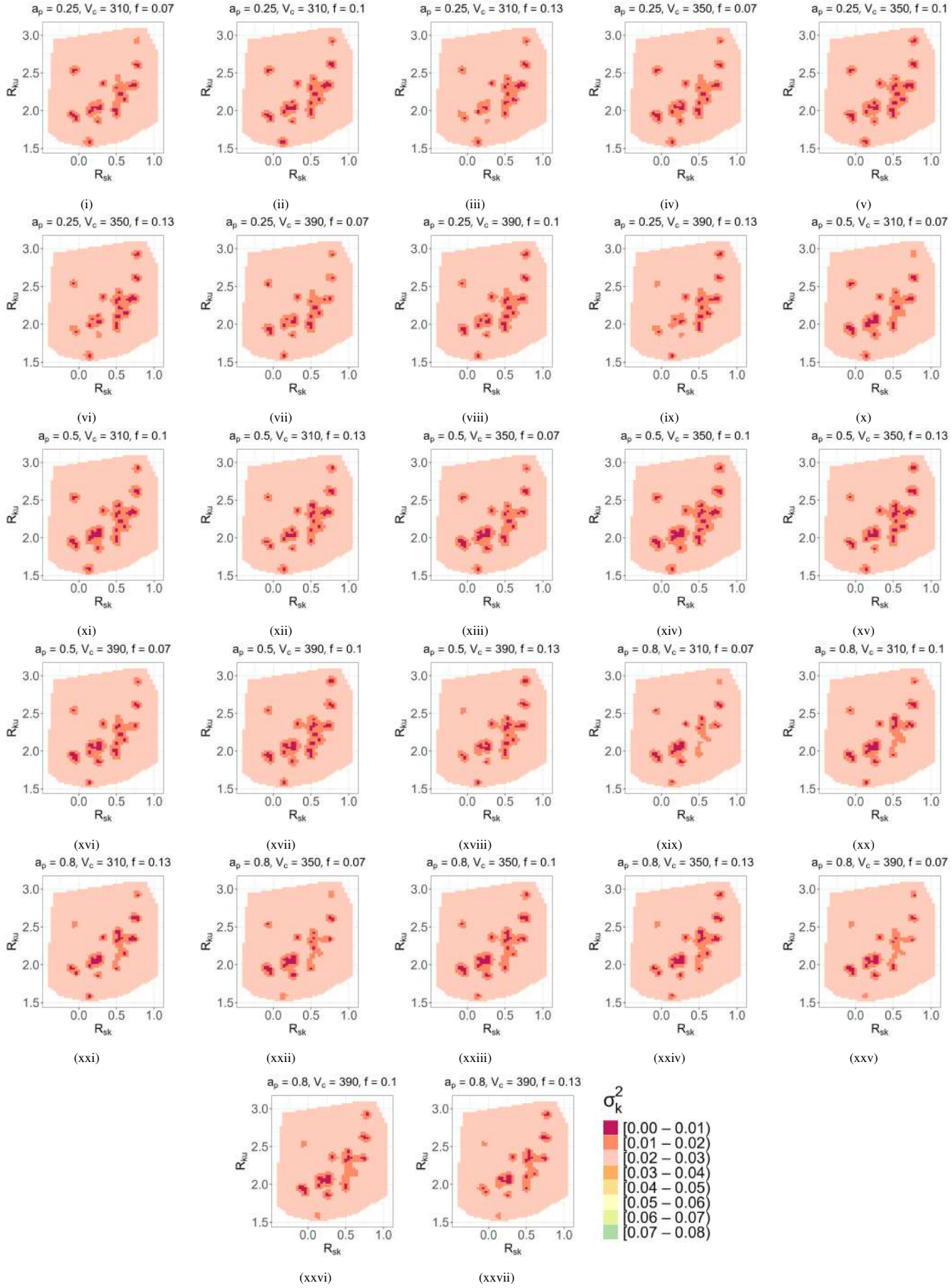


Figure S8: Spatial distribution of kriging prediction variance (σ_k^2) corresponding to *MLE-based* universal kriging estimates of R_a for all 27 machining parameter combinations (a_p–V_c–f).